



D(2nd Sm.)-Computer Science-H/DSCC-2/CCF

2025

COMPUTER SCIENCE — HONOURS

Paper : DSCC-2

(Problem Solving Using C)

Full Marks : 75

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer **question no. 1** and **any three** from **Section - A** and **any five** from **Section - B**.

1. Answer **any five** questions :

2×5

- State the difference between `fwrite()` and `fprintf()`.
- What is the use of `feof()` and `ferror()` in file handling?
- Mention the purpose of the `break` and `continue` statements.
- Define an array. How is it different from a variable?
- What is a function prototype? Why is it important?
- Briefly describe the main symbols used in a flowchart.
- How does 'sizeof' operator work? Explain.
- Swap two integers `p` and `q` using bitwise operators only.

Section - A

2. Write a C program to read a string from the user and count the number of vowels.

5

3. (a) Differentiate between `malloc()` and `calloc()` functions.

(b) Write a C program to dynamically allocate memory for :

- One structure variable, and
- An array of three structure variables

The structure should contain roll and age as members. Use appropriate memory allocation functions.

2+3

4. How does a 'switch' differ from the 'if' statement? Explain with example.

5

5. Compare parameter passing techniques of 'call by value' and 'call by address' with an example.

5

Please Turn Over

(3455)

6. Write a program in C to create a function (using pointer parameters) that compares two integer arrays to check whether they are identical. The function returns 1 if identical, 0 otherwise. 5

Section - B

7. (a) Discuss about the following operators in C, with example :
 (i) Bitwise operators
 (ii) Increment and decrement operators.
 (b) Write a C program to check whether a number is prime or not. 4+6
8. (a) What is a ternary operator in C? Write a C program to find the largest of three integers using only the ternary operator.
 (b) Write a program in C that reads a positive integer and outputs the number of 1's in its binary representation. 5+5
9. (a) What is a macro in C? Compare and contrast macros and functions with suitable examples.
 (b) What is type casting? Explain implicit and explicit type casting with examples.
 (c) Which functions are used to compare two strings in a case-sensitive and a case-insensitive manner? (1+3)+(1+3)+2
10. (a) What is a void pointer in C? Why is it used?
 (b) Differentiate between a structure and a union in C.
 (c) Briefly describe the different storage classes available in the C language. (1+2)+3+4
11. What will be the output of the following C program codes? Justify your answers.
 (Assume all necessary header files such as `#include <stdio.h>` are included.)
- (a) `#define SQUARE(x) x * x`
`int main() {`
`int a = 5, b = 3;`
`printf("%d\n", SQUARE(a + b));`
`return 0; }`
- (b) `int main() {`
`int arr[] = {1, 2, 3, 4};`
`int *ptr = arr;`
`printf("%lu\n", sizeof(arr) / sizeof(ptr));`
`return 0; }`
- (c) Write a C program to find the sum of digits of a 5 digit number using recursion. 2+2+6

12. (a) Explain the actual and formal arguments of a function, using a suitable example.
 (b) What is the advantage of using static variables inside a function? Explain with an example.
 (c) Explain the difference between & and * operators in the context of pointers. 5+3+2
13. (a) How can we return multiple values from a function in C? Illustrate with an example.
 (b) Write a code snippet in C to allocate a block of memory using malloc().
 (c) Write the syntax for opening a file with various modes and closing a file. 5+3+2
14. (a) What are the problems faced while using array in a program?
 (b) Write down what will be the output of the following function :

```
void abc(char *s)
{
    if(s[0]=='\0') return;
    abc(s+1);
    abc(s+1);
    printf("%c",s[0]);
}
main()
{
    abc("123");
}
```

- (c) What will be the output of the following code?

```
#include <stdio.h>
typedef struct
{
    int A;
    char B;
    char C;
} InfoData;
int main(int argc, char *argv[])
{
    printf("\n Size of Structure = %d\n\n", sizeof(InfoData));
    return 0;
}
```

2+4+4